

COURSE DETAILS

Faculty	: Information and Communication Technology
Programme	: Degree in Software Engineering with Multimedia
Year/ Semester	: Year 3 Semester 2
Lecturer	: Mr Tlali Mohale
Principal Lecturer	: Mrs Mareitumetse Macheli
Pre-requisite	: Software Requirement Engineering
Contacts hours	: 3 hours
Methods of Delivery	: Lectures and Tutorial sessions
Credits	: 10
Contact hours	: 3 (2 hours lecture +1 hour tutorial)
No. of weeks	: 17 weeks long semester (12 teaching, 1 midterm, 1 non-contact week, 1 revision, 2 Final Exams)
Venue	: Room 1
Day	: Monday
Time	: 08:30 – 10:30

Prepared by : MR TLALI MOHALE

Checked by: MRS MAREITUMETSE MACHELI

Signature :

Date: 23/01/2026

Signature:

Date: 23/01/2026

This document comprises the following:

- Essential Information
- Specific Course Information
- Course Rules & Regulations
- Grades
- Plagiarism
- Course Introduction
- Course Aims & Objectives
- Learning Outcome
- Specific Generic Learning Skills
- Syllabus + Lecture Outline
- References
- Assignment Schedule
- Assessment Criteria
- Specific Criteria

Other documents as follows will be issued to you on an ongoing basis throughout the semester:

- Handouts for Assignments
- Submission Requirements + Guidelines

1.0 ESSENTIAL INFORMATION

- All courses other than electives are **'significant courses'**

- As an indicator of workload one credit carries and additional 2 hours of self study per week. For example, a course worth 3 credits require that the student spends an additional 6 hours per week, either reading, completing the assignment or doing self directed research for that course.
- Submission of ALL assignment work is compulsory in this course failure to do so, a DNS (Did not submit) grade will be awarded. An overall grade of DNC(Did not complete)will be awarded for those who fail to submit a major piece of assessment work(major assignment) or sit for either the midterm examination or final examination. A student cannot pass this course without having to submit ALL assignment work by the due date or an approved extension of that date.
- Submission of ALL assignment work is compulsory in this course. A student cannot pass this course without having to submit ALL assignment work by the due date or an approved extension of that date.
- All assignments are to be handed on time on the due date. Students will be penalised 10 percent for the first day and 5 percent per day thereafter for late submission (a weekend or a public holiday counts as one day). Late submission, after the date of Board of Studies meeting will not be accepted.
- Due dates, compulsory assignment requirements and submission requirements may only be altered with the consent of the majority of students enrolled in this course at the beginning/early in the program.
- Extensions of time for submission of assignment work may be granted if the application for extension is accompanied by a medical certificate.
- Overseas travel is not an acceptable reason for seeking a change in the examination schedule.
- Only the Head of School can grant approval for extension of submission beyond the assignment deadline.
- Re-submission of work can only receive a 50% maximum pass rate.
- Supplementary exams can only be granted if the level of work is satisfactory **AND** the semester work has been completed.
- Harvard referencing and plagiarism policy will apply on all written assignments.

2.0 SPECIFIC COURSE INFORMATION

- Attendance rate of 80% is mandatory for passing course.
- All grades are subject to attendance and participation.
- Absenteeism at any scheduled presentations will result in zero mark for that presentation.
- Visual presentation work in drawn and model form must be the original work of the student.
- The attached semester program is subject to change at short notice.

3.0 COURSE RULES AND REGULATIONS:

Assessment procedure:

- These rules and regulations are to be read in conjunction with the UNIT AIMS AND OBJECTIVES
- All assignments/projects must be completed and presented for marking by the due date.
- Marks will be deducted for late work and invalid reasons.
- All assignments must be delivered by the student in person to the lecturer concerned. No other lecturer is allowed to accept students' assignments.
- All tests/examinations are compulsory.
- Students must sit the test/examination on the notified date.
- Students are expected to familiarise themselves with the test/examination timetable.
- Students who miss a test/examination will not be allowed to pass.
- Any scheduling of tutorials, both during or after lecture hours, is TOTALLY the responsibility of each student. Appointments are to be proposed, arranged, confirmed, and kept, by each student. Failure to do so in a professional manner may result in penalty of grades. Tutorials WITHOUT appointments will also NOT be entertained.
- Note that every assignment is given an ample time frame for completion. This, together with advanced information pertaining deadlines gives you NO EXCUSE not to submit assignments on time.

4.0 GRADES

All courses and assessable projects will be graded according to the following system. With respect to those units that are designated 'Approved for Pass/Fail' the grade will be either PA or F:

<i>Grade</i>	<i>Numeric Grade</i>	<i>Description</i>
90 – 100	A+	Pass with Distinction
85 – 89	A	
80 – 84	A-	
75 – 79	B+	Pass with Credit
70 – 74	B	
65 – 69	B-	
60 – 64	C+	Pass
55 – 59	C	
50 – 54	C-	
47 – 49	PP	Supplement
0 – 46	F	Fail

EXP	Exempted
PP	Pass Provisional with extra work needed
X	Ineligible for assessment due to unsatisfactory attendance
D	Deferred

W	Withdraw
DNA	Did Not Attend Course
DNC	Did Not Complete Course

5.0 PLAGIARISM, COPYRIGHT, PATENTS, OWNERSHIP OF WORK: STUDENT MAJOR PROJECT, THESES & WORKS

5.0.1 Academic Dishonesty

Academic Dishonesty or academic misconduct is any type of cheating that occurs in relation to a formal academic exercise. It can include:

Plagiarism: The adoption or reproduction of original creations of another author (person, collective, organization, community or other type of author, including anonymous authors) without due acknowledgement.

Fabrication: The falsification of data, information or citations in any formal academic exercise.

Deception: Providing false information to an instructor concerning a formal academic exercise – e.g. giving a false excuse for missing a deadline or falsely claiming to have submitted work

Cheating: Any attempt to give obtain assistance in a formal academic exercise (like an examination) without due acknowledgement.

Bribery: or paid services. Giving certain test answers for money.

Sabotage: Acting to prevent others from completing their work. This includes cutting pages out of library books or wilfully disrupting the experiments of others.

Students involved in academic dishonesty will be treated as follows:

1. Plagiarism and fabrication in Assignments, Projects or Presentations: A zero (0) mark will be awarded to the work.
2. Deception: 10 % will be deducted on the first day of late submission and 5% a day thereafter. Late submission on or after BOS will not be accepted.
3. Cheating: Any form of assistance in a test will be rewarded a zero (0) mark. While in an examination the student automatically repeats the module.
4. Bribery, Sabotage or Professional Misconduct: Committing any of these offenses will lead to discontinuity from the University for a certain period depending on the seriousness of the offence.

5.0.2 Copyright, Patents and Ownership of Work

All use of the borrowed work, whether text or images must be referenced and acknowledged. Limkokwing is not liable for any infringement of copyright within students' work. Students are to abide by International Copyright laws. Students engaged in a project activities will be required, as a condition of acceptance as a student, to agree to assign to Limkokwing their right, title and interest in any invention or visual work arising from their studies with the University. Limkokwing has the right to keep students work for use in archives, exhibitions and events. Where students wish to keep original work they must apply in writing and gain permission from the Academic Management.

6.0 Course INTRODUCTION

This course is about study of fundamental design concepts, design notations, and architectural design methods for large-scale software systems. Several design methods are presented and compared, with examples of their use. Concepts such as information hiding data abstraction, concurrency, and object-oriented software construction are discussed in depth. Students participate in a group project on software design.

7.0 Course AIMS AND OBJECTIVES

- Introduce fundamental design concepts and design notations
- Discuss different life cycle model and their impact on software development.
- Increase understanding of several design methods that are presented and compared, with examples of their use.
- To identify the design of a relatively complex software system.

8.0 LEARNING OUTCOME

Upon completion of this course, participants should know and understand

- Software requirements and its role in determining the architecture design
- Object Oriented Architecture (OO Analysis and Design)
- Design Principles and their role in leading to sustainable and dependable system
- The domain of application of different architectural styles
- Model Driven Software Development
- Software Product Families and Feature Modeling

9.0 SPECIFIC GENERIC LEARNING SKILLS

Upon completion of the Course, student will acquire skills in:

- Design Concept
- Understanding an Object Oriented Analysis and modeling
- Object Oriented Design Method

10.0 PROGRESSION POLICY

Students can only enroll into a higher semester if they have passed all courses in the preceding semester or have failed not more than two courses. A student cannot register for any course at a higher level if they fail its lower unit which in essence is a pre-requisite. If student CGPA is less than 2.0 could not proceed to the next semester.

11.0 UNIT SYLLABUS + LECTURE OUTLINE:

Week 1

Lecture 1 Introduction

Lecture Synopsis

- Introduction
- Aims and Outcomes
- Teaching and Assessment
- Software Development Terminology
- Design Objectives and Design Problems
- Why Do Things Go Wrong?
- Project Life Cycles

Handouts: Notes

Week 2

Lecture 2 Software Development

Lecture Synopsis

- Software Development Terminology
- Design Objectives and Design Problems
- Why Do Things Go Wrong?
- Project Life Cycles

Handouts: Notes

Week 3

Lecture 3 Object Orientation

Lecture Synopsis

- What is object-orientation?
- Why object-orientation?
- Objects concepts
- Summary

Handouts: Notes

Week 4

Lecture 4 Requirements and Use Cases

Lecture Synopsis

- Requirements capture
- Requirements modelling
- Use-cases
- UML Use-Case Diagrams
- Background to UML

Handouts: Notes

Week 5

Test 1

Lecture 5 An Overview of UML

Lecture Synopsis

- Background to UML
- Models and Diagrams
- UML Tools

Handouts: Notes

Week 6

Lecture 6 Class Models and Diagrams

Lecture Synopsis

- Class models
 - Finding classes and associations
 - Basics of UML Class Diagrams
 - ATM Case Study: initial class model
 - More about associations
 - More about attributes and operations
 - ATM Case Study
-

Handouts: Notes

Week 7

- **Midterm Examination**
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Week 8

- **None Contact week**
-

Week 9

Lecture 7 Collaboration and Interactions

Lecture Synopsis

- Views and Diagrams
- Responsibility-Driven Design (RDD)
- Collaboration
- Collaboration and Interaction
- Object stereotypes
- Basics of Interaction Sequence Diagrams
- ATM Case Study: the withdrawal use-case
- ATM Case Study: the withdrawal use-case

Handouts: Notes

Week 10

Lecture 8 Object Behaviour

Lecture Synopsis

- The Behavioural view
- Basics of UML Statechart Diagrams
- ATM Case Study
- Activity Diagrams

Handouts : Notes

Week 11

Lecture 9 UML Notations

Lecture Synopsis

- Activity Diagrams
- ATM Case Study: withdrawal use-case
- Packages and Subsystems
- Implementation Diagrams (very briefly)

Handouts : Notes

Week 12

Lecture 10 Design Quality

Lecture Synopsis

- What is Quality?
- Quality in Software
- Key design objectives:
- Measurable software attributes:
- Product and Process

Handouts: Notes

Week 13

Lecture 11 Product Quality

Lecture Synopsis

- Verification, validation and testing
- ATM Case Study

Handouts: Notes

Week 14

Lecture 12 Software Patterns

Lecture Synopsis

- What are Patterns
- Design patterns
- Using Design Patterns

Handouts: Notes

12.0 REFERENCES

Recommended Book

- John Ousterhout, A Philosophy of Software Design, Yaknyam Press; 1st edition, 2018
- T.C.Lethbridge and R.Laganiere, Objec-Oriented Software Engineering, 2nd Edition: Practical Software Development using UML and Java, McGraw-Hill, 2005

Additional Books

- Mark Richards, Pramod Sadalage, Zhamak Dehghani, Software Architecture: The Hard Parts: Modern Trade-Off Analyses for Distributed Architectures 1st Edition Neal Ford, 2021
- Favre, Lilliana, UML AND THE UNIFIED PROCESS, Idea Group Inc., 2003
- Yang, Hongji, Software Evolution with UML and XML, Idea Group Publishing, 2004
- Eric J. Braude, Software Design: From Programming to Architecture, John Wiley & Sons, 2004, ISBN: 0 - 471- 42920 -1.
- Pressman, R.S. Software Engineering: A Practitioner's Approach, 6th Ed., McGraw-Hill Companies, Inc. 2005.
- Somerville, Ian. Software Engineering, 6th Ed., Addison-Wesley, 2001.
- David Budgen, Software Design, Pearson, Addison Wesley, second edition 2003

13.0 ASSESSMENT SCHEDULE

Assignment description	issue date	due date	%
Test	WEEK4	WEEK4	10%
Group ASSIGNMENT	Week 1	Week 12	25%
MID SEMESTER TEST	week 7	week 7	25%
Final examination :	week 15	week 15	40%
Total			100%

14.0 ASSESSMENT CRITERIA

Process of grading and criteria used to determine the grades, passes and high distinctions.

15.0 Specific Criteria

- Each assignment will be handed out with the project brief and will vary, depending on the teaching and learning objectives of the specific assignment.
- Each student will receive a completed assessment sheet back with their marks, thereby giving student feedback on each set criterion and the project as a whole.
- Marks for each project will be posted on the Bulletin Board with student number within 2 weeks of hand-in date.

